Advanced analysis
Antimicrobial resistance prediction

Advanced analysis appendix | Eight figures with linked result tables

Validation rows	Test rows	New uncertainty checks
36,291	37,175	500 draws
3,147 patients	3,203 patients	Patient bootstrap; model comparison uses 200

Prior history carries useful added information. Validation resistance AP is 0.250 with core fields, 0.375 with prior history, and 0.373 with all features. The paired comparison shows no clear extra gain from labs and vital signs under this recipe.

One cutoff behaves very differently across drugs. It misses all 108 resistant piperacillin/tazobactam rows and flags all ampicillin rows. Good overall scores do not remove this problem.

Calibration reduces probability error. Test Brier falls from 0.179 to 0.118. Drug-level checks still matter, especially where there are few resistant patients.

More recall brings more false alerts. Moving from the original cutoff to the 90% validation-recall target reduces misses from 49 to 18 per 1,000 rows, while false alerts rise from 222 to 438.

Missing values and missing labels limit the claims. Availability patterns contain useful information, including recorded prior resistance. Missing outcomes remain unknown. Label coverage is conditional on inclusion in the analytical cohort.

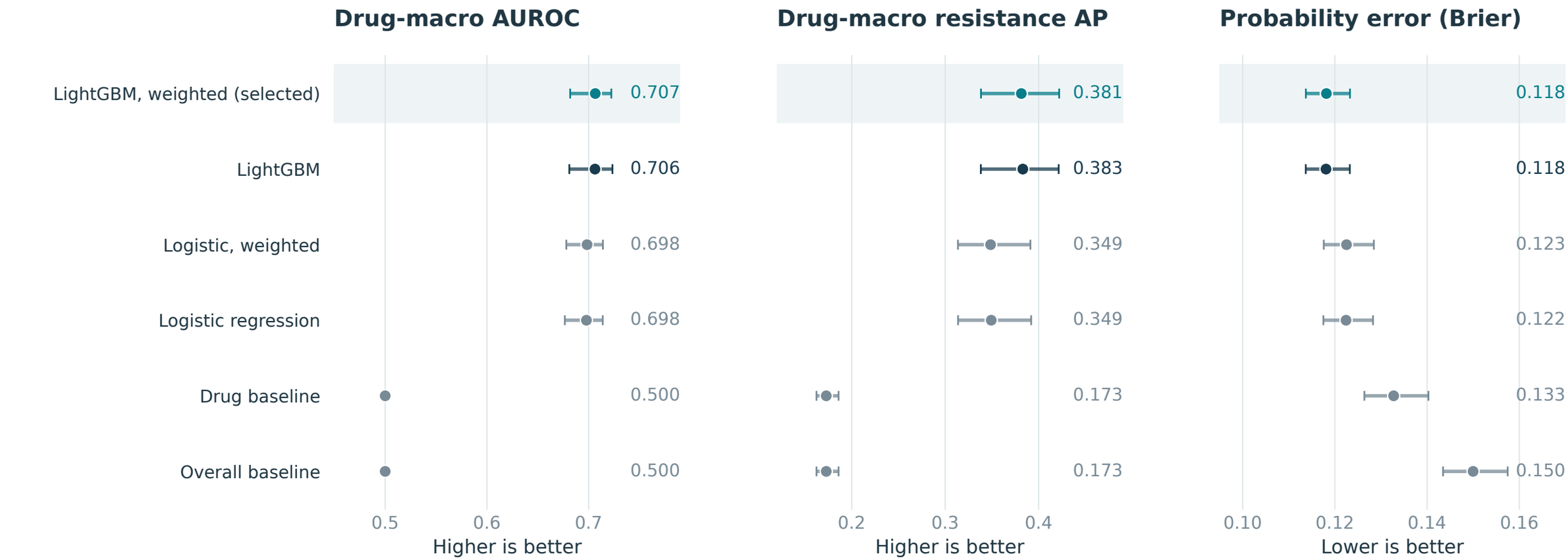
Reading guide: 01 models; 02 feature groups; 03 drug performance; 04 overall calibration; 05 calibration by drug; 06 cutoff tradeoffs; 07 label coverage; 08 missingness.

The feature comparisons use validation only. The other model checks describe the already examined test set. Treat every new finding as exploratory.

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Model differences, with uncertainty

Original test results | six comparisons | the selected model stays fixed



Dots show estimates; lines show 95% patient-bootstrap intervals from the primary run (200 draws). AP is average precision for resistance. Weighted and unweighted LightGBM are very close; this is not evidence of a clear benefit from class weighting.

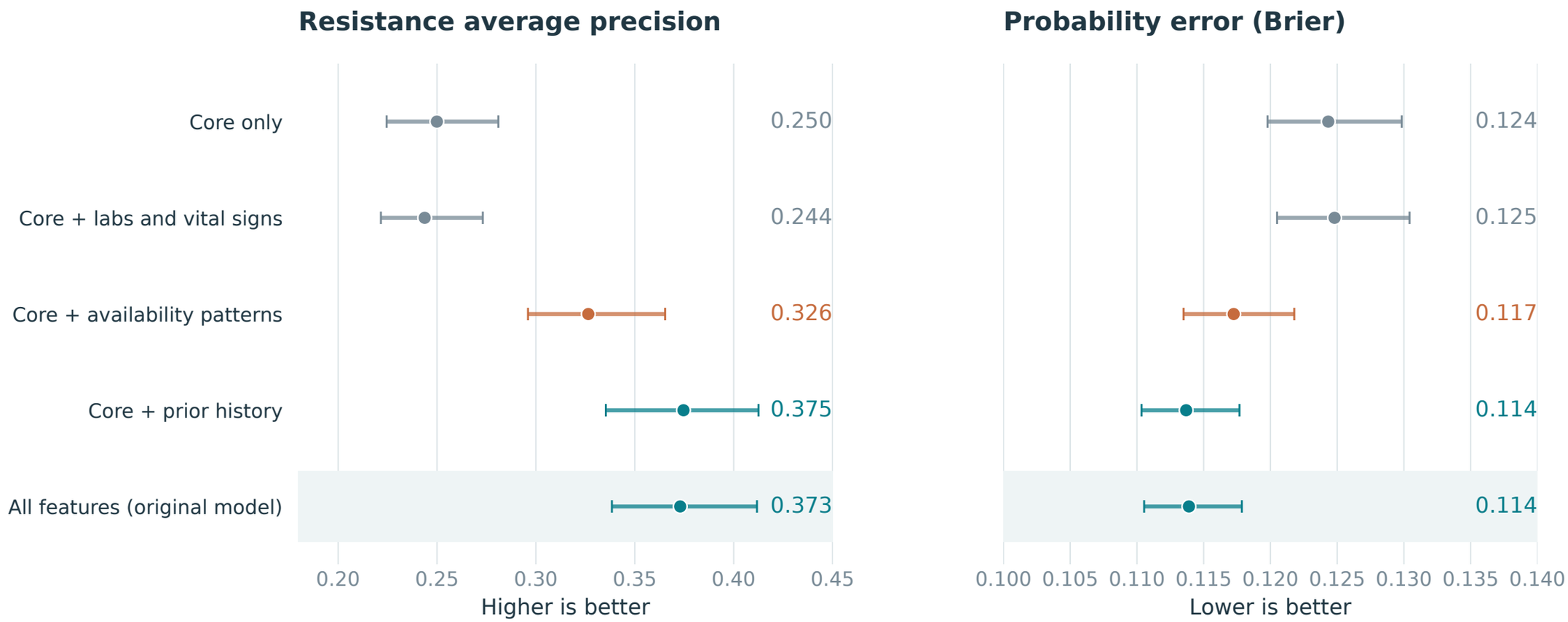
The selected weighted LightGBM has average resistance AP 0.381 across drugs. The unweighted version scores 0.383. The difference is small; these results do not show a clear benefit from class weighting.

AUROC checks ranking. Resistance average precision (AP) checks how well resistant results rise to the top. These two scores give each drug equal weight. Brier averages probability error over candidate rows. All models and the selected model are unchanged from the original analysis.

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What each feature group adds on validation

Feature-group comparisons on validation only | identical patient partitions and fixed model settings



Core = drug, age band, gender code, culture type and ward flags. The availability control keeps missing-value patterns but removes recorded numeric values. Lines are 95% patient-bootstrap intervals (500 shared draws). This experiment is exploratory and does not measure causal effects.

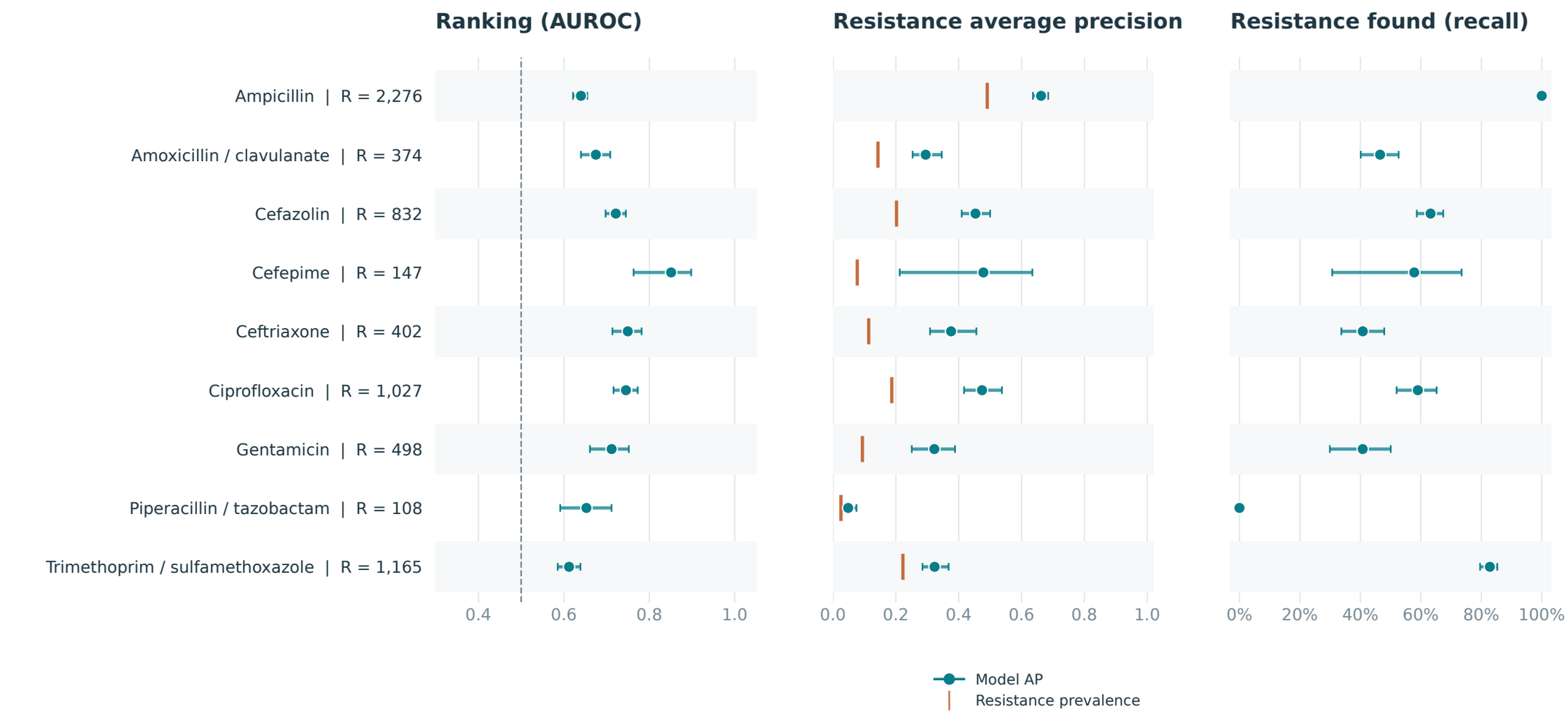
Core plus prior history scores 0.375 for resistance AP; all features score 0.373. The paired full-minus-history difference is -0.002 (95% interval -0.011 to 0.009). This fixed recipe shows no clear added improvement from labs and vital signs once prior history is included. Similar scores do not prove equivalence.

Core plus availability patterns scores 0.326. The full model scores 0.046 AP higher (interval 0.029 to 0.060). The control keeps missing values missing and replaces observed numbers with a constant. A recorded prior-resistance entry already conveys history even when its time lag is erased, so this signal is not only about documentation habits.

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Performance varies substantially across antibiotics

Frozen selected model | test results | confidence intervals keep each patient's records together



R counts resistant test rows. Lines are 95% patient-bootstrap intervals (500 draws). Orange ticks mark resistance prevalence, a no-skill AP baseline. Recall uses the original cutoff. Boundary intervals at 0% or 100% do not imply certainty for future patients. AP also depends on prevalence and case mix.

The same global cutoff finds none of the 108 resistant piperacillin/tazobactam rows. It flags just 1 row, which is susceptible. For ampicillin it finds every resistant row, but flags all 4,637 rows. Overall recall of 73.2% hides these large differences.

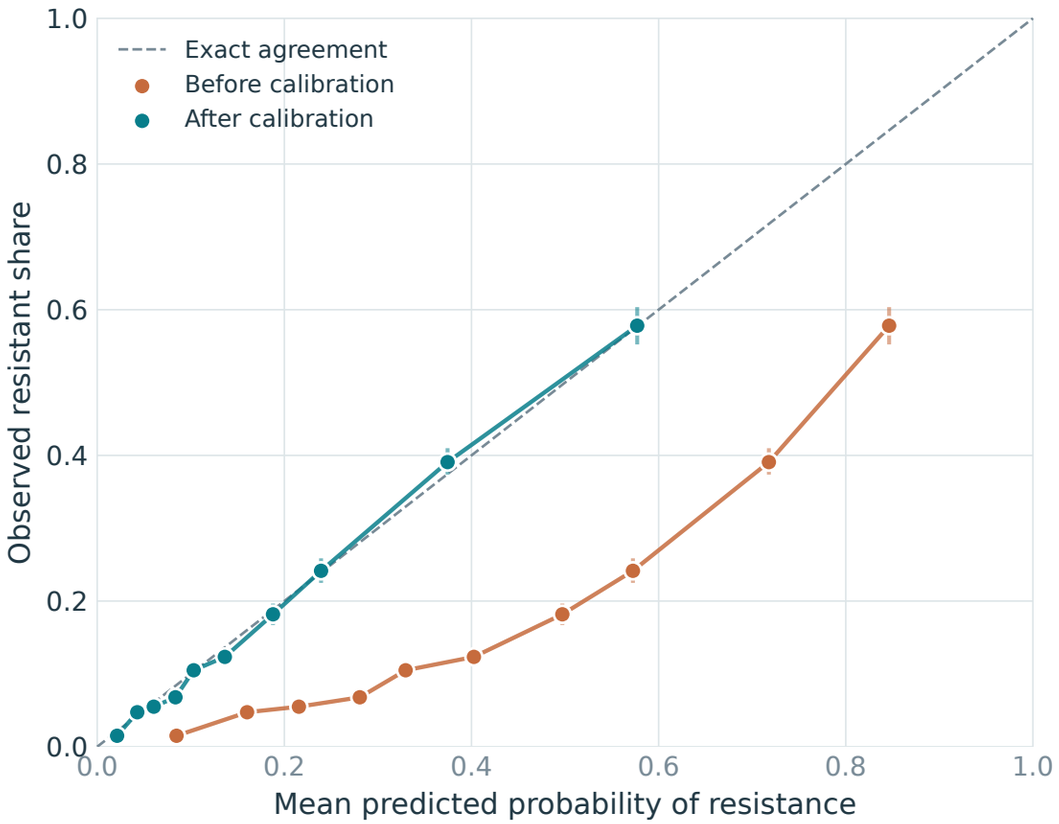
Ranking and cutoff performance answer different questions. Drug-level AP also changes with resistance prevalence and case mix. Wide intervals, such as those for cefepime, show that many rows can still represent relatively few patients with resistant results. A bootstrap interval at zero or one is not certainty about future patients.

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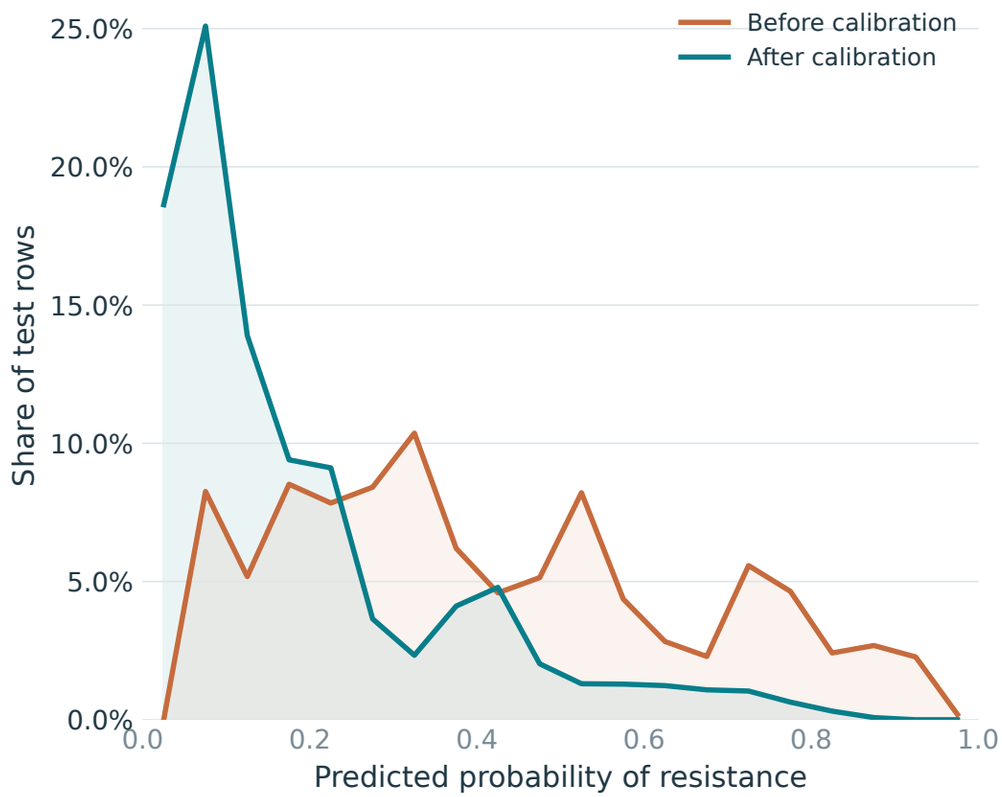
Probability checks before and after calibration

Resistance probability = 1 minus predicted susceptibility | calibration was fitted on training patients only

Agreement with observed outcomes



Where the predictions fall



Reliability-bin edges were fixed from validation predictions. Vertical lines show 95% patient-bootstrap intervals for the observed resistant share.
Distribution lines summarize fixed-width probability bins. These are exploratory checks on the existing test set, not a new calibration fit.

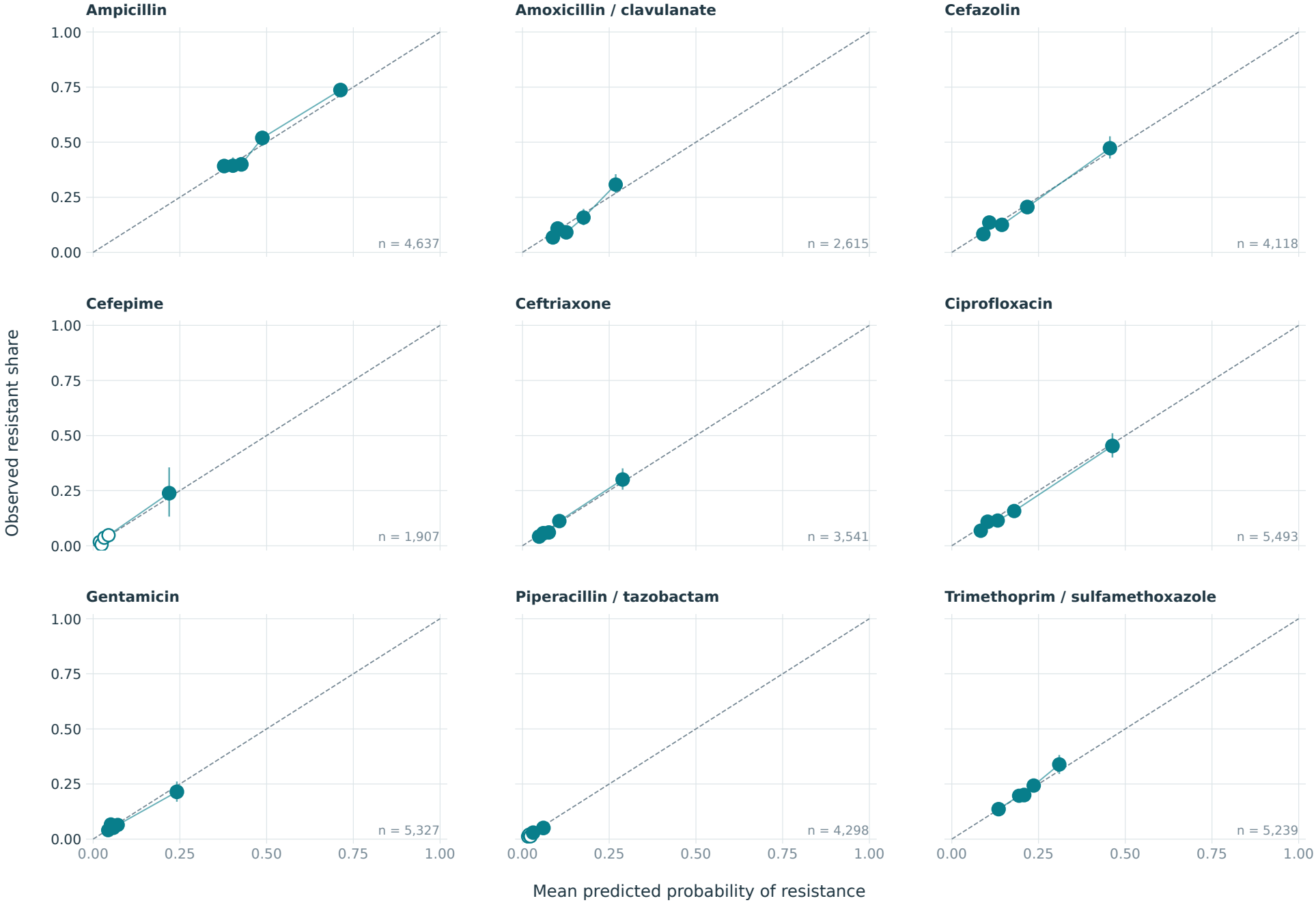
Brier falls from 0.179 before calibration to 0.118 after calibration. The paired decrease is 0.061 (95% interval 0.057 to 0.065). The calibrator was fitted on held-out training patients in the original run, not on these test outcomes.

The mean calibrated resistance probability is 18.6%; the observed resistant share is 18.4%. The reliability curve checks agreement across probability ranges. The lower Brier score is useful, but by itself does not prove good calibration at every probability.

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A good pooled calibration curve can hide drug differences

Selected model | test outcomes in validation-defined bins | open circles mark sparse bins



The dashed line marks exact agreement. Larger points mark larger bins; lines show 95% patient-bootstrap intervals (500 draws). Sparse means fewer than 100 rows or fewer than 20 resistant results in a bin. Intervals are pointwise and do not prove calibration across all probabilities.

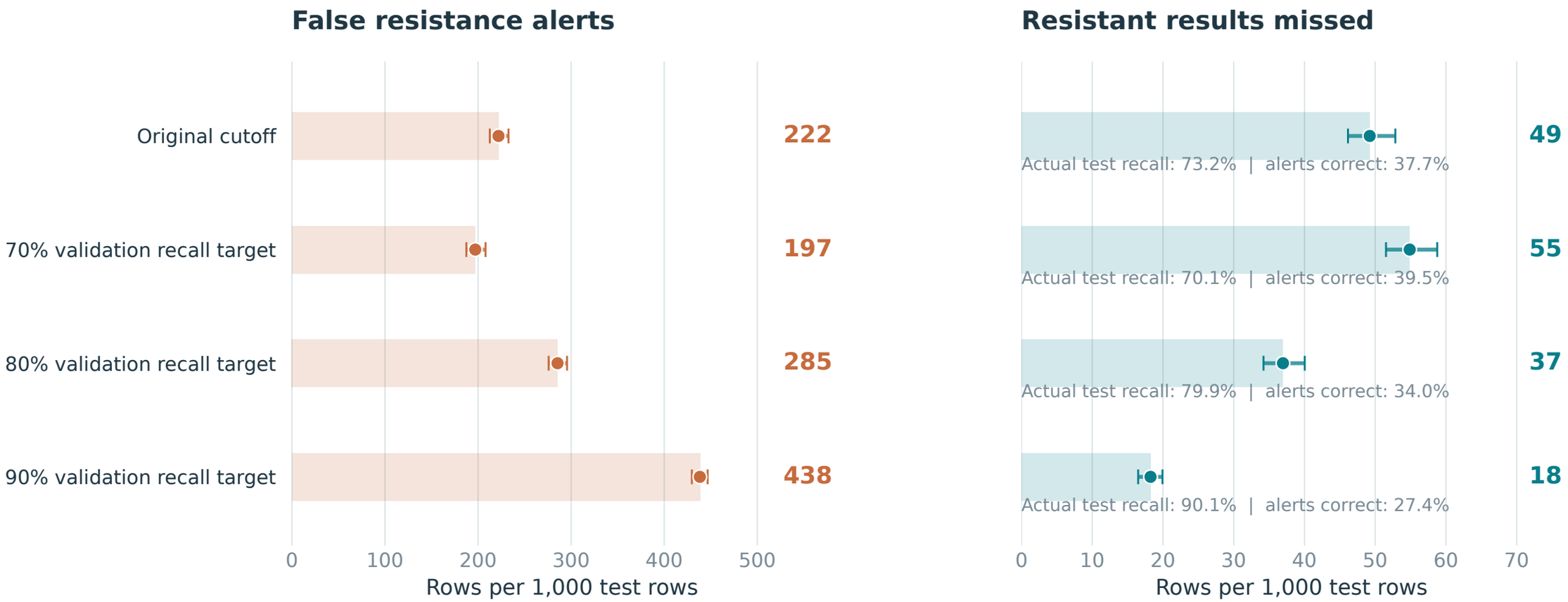
The drug panels show where probability checks have enough support and where they do not. Bin boundaries came from validation predictions. Open circles mark fewer than 100 test rows or fewer than 20 resistant results in a bin. Large uncertainty in a sparse bin should remain visible.

An acceptable pooled curve can conceal drug-level errors. Read each panel with its counts and intervals. These are descriptive checks on the existing test set. No drug-specific recalibration was fitted, and the plots do not show that the model will be calibrated at another hospital.

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Finding more resistance brings more false alerts

Cutoffs chosen on validation, then applied unchanged to test | counts per 1,000 candidate test rows



False alerts are susceptible rows flagged as resistant. Misses are resistant rows flagged as susceptible. Lines are 95% patient-bootstrap intervals.
Targets describe validation recall; actual test recall may differ. These retrospective operating points are not clinical treatment rules.

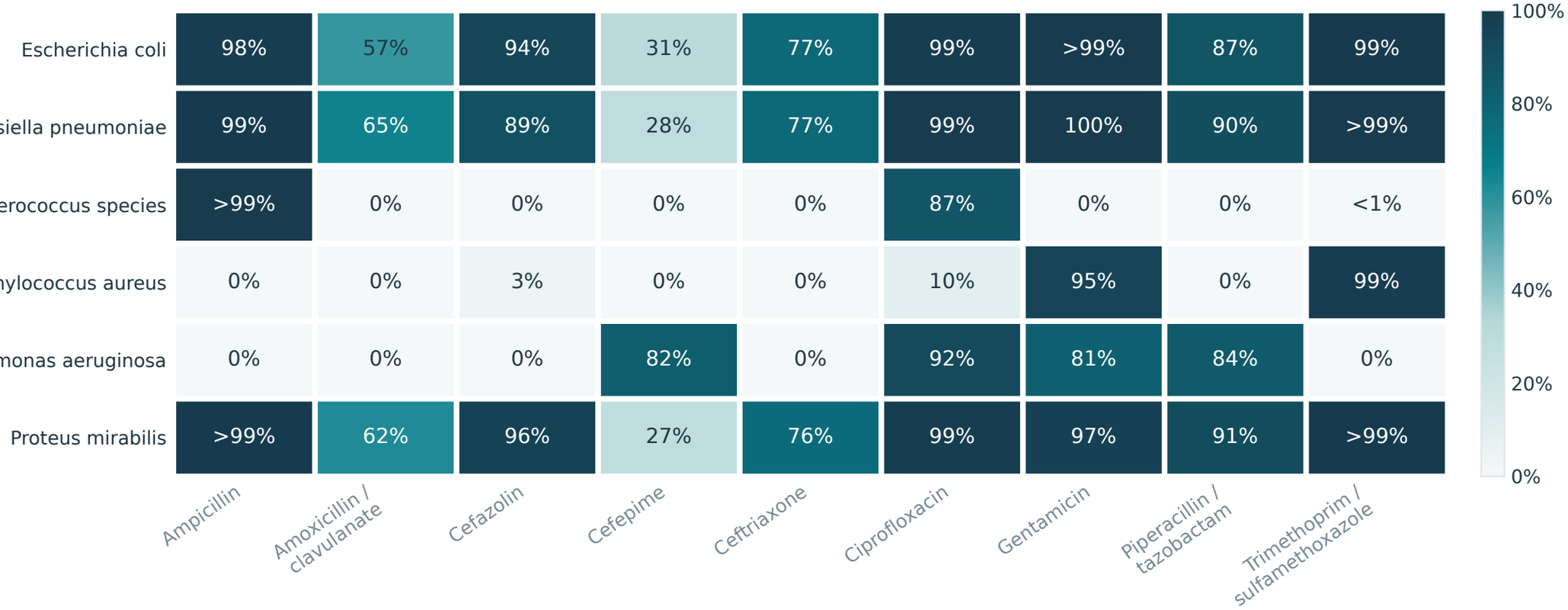
The original cutoff gives about 222 false resistance alerts and 49 missed resistant results per 1,000 test rows. The 90% validation-recall target gives about 438 false alerts and 18 misses. Its actual test recall is 90.1%, while only 27.4% of alerts are correct.

Each new cutoff was chosen on validation and then held fixed for test. A pooled recall target does not ensure that each drug reaches that target. These counts describe classification errors per candidate row, not per patient. They are not treatment rules or estimates of clinical benefit.

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The available labels are uneven across organisms and drugs

Six most common training organisms | test-set retained label coverage | empty outcomes remain unknown



Each percentage is the share of included organism-events with a clear S/R label for that drug. The denominator includes events with at least one retained panel result. This is not the raw testing rate. A zero means no retained label, not susceptibility or resistance.

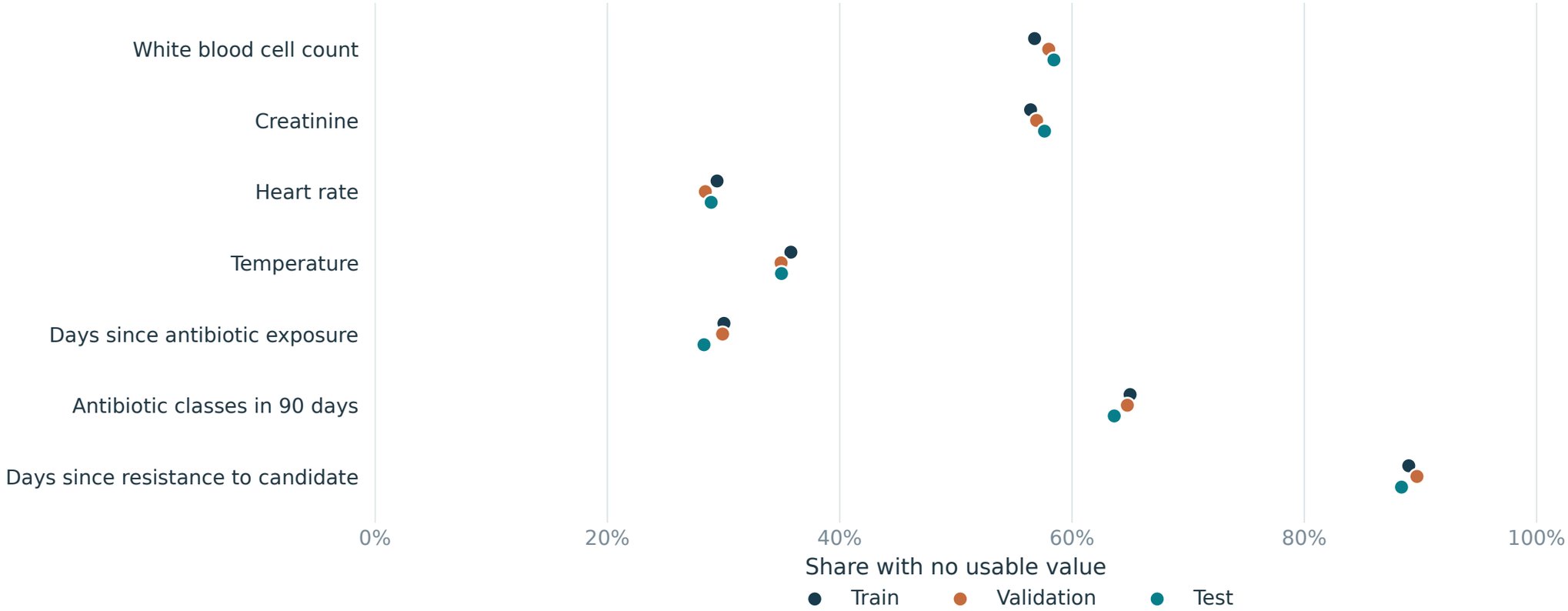
Among included E. coli events, 30.7% have a retained cefepime label, compared with 99.7% for gentamicin. The six organisms were chosen by training frequency, rather than by test performance.

The denominator contains organism-events with at least one clear, retained susceptible or resistant result in the chosen panel. This is coverage within the analytical data, not the raw testing rate. A blank or zero does not mean susceptibility or resistance. The data cannot establish why an outcome is absent or how the model would perform on unobserved combinations.

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Missing information is common in every data split

Source information before imputation | each culture counted once for the shared clinical fields



Laboratory, vital-sign and antibiotic-exposure fields use one record per culture. Candidate resistance history uses one record per culture and candidate drug. No record means unknown; it does not establish no prior exposure or no prior resistance. Similar missing rates alone do not establish comparable patients.

In test data, white blood cell count is missing for 58.4% of cultures. Prior candidate-resistance history is missing for 88.4% of culture-candidate pairs. A missing history record cannot establish that a patient had no past exposure or resistance.

Shared clinical fields are counted once per culture. Candidate resistance history is counted once per culture and candidate. This avoids giving a culture extra weight just because it has more organism or drug rows. Similar missing rates across splits do not prove similar case mix or complete source records.

What these results can support

Status and scope. This appendix is an exploratory follow-up after the original test results were seen. It sits outside the original five-page report body. The original selected weighted LightGBM model, its predictions and its primary test scores remain fixed. No new winning model was selected.

Feature comparisons. Four reduced models use the original proper-fit patients, held-out training calibration patients, balanced training weights and fixed LightGBM settings. Preprocessing is fitted only on proper-fit data. Reduced models are scored on validation only. Those validation patients had already informed the original model choice, so these are not fresh independent estimates.

Uncertainty. New intervals use 500 shared patient-bootstrap draws. All rows for a sampled patient stay together. Paired comparisons use the same draws on both sides. The original model-comparison figure retains its 200-draw intervals. Intervals are pointwise, not adjusted for all comparisons, and do not include model-training or earlier selection uncertainty. Boundary intervals can collapse even when future performance is uncertain.

Calibration and cutoffs. Probabilities refer to resistance, P(R) = 1 - P(S). Calibration was fitted on training patients. Reliability-bin boundaries and new recall-target cutoffs use validation predictions. They are then kept fixed for test. No test-fitted calibrator or threshold is introduced.

Units and labels. The prediction unit is one patient, culture, organism and candidate drug. Labels are susceptible = 1 and resistant = 0. Intermediate, unknown and conflicting labels stay excluded. Error counts refer to candidate rows. Coverage refers only to organism-events with at least one retained panel label. Missingness uses culture or culture-candidate counts as stated in each table.

Limits that still apply. The nine-drug list remains provisional until the instructor confirms it. The supplied tables do not include a hospital identifier, so hospital validation is unavailable. Shifted dates do not support a calendar trend claim. No feature comparison establishes causality, no threshold establishes treatment benefit, and no result establishes performance at a new site.

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